Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/09/2026
PI Name:	Biga, Peggy		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Peggy.Biga@txstate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	Texas State University		
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PI Web Page:			
City:	San Marcos	**State:**	TX
Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	08/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rabenberg, Ellen	**Contact Phone:**	(256) 544-7264
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	During the remainder of 2025-26, Peggy Biga (Biology Chair, Texas State University) will be serving as interim Principal Investigator (PI) due to Texas State University policy upon the formal retirement of Robert McLean. Dr. McLean will still be involved and will be responsible for the scientific completion, manuscript writing, and any necessary reports. Mark Ott has retired from NASA Johnson Space Center; however, he continues on the project team (Ed., 4/9/26).		
COI Name (Institution):	Barrila, Jennifer (Arizona State University) Ott, C. Mark (NASA Johnson Space Center - Retired) Nickerson, Cheryl (Arizona State University)		
Grant/Contract No.:	80NSSC22K1361 ; NNX17AC79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. As a result of supplemental funding, we are now able to gather genetic data on the mechanisms by which mixed bacterial species are able to form biofilms in microgravity conditions. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	With the retirement of Robert McLean, the new Principal Investigator (PI) for this project is Peggy Biga. Biology Chair for Texas State University.
Bibliography Type:	Description: (Last Updated:)